Victor Chua meow

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Singapore Citizen | linkedin.com/in/vicchua123 | Available part-time, 15–20 hrs/week from August 2026

AI/LLM Solutions | On-Premises & Localised Deployment | Evaluation & Internal Enablement

Technical Skills

AI & LLM Systems
– **RAG**, **ChromaDB**, Vector Embeddings, Semantic Chunking, Re-ranking, Metadata Filtering
– **On-Premises Inference**, **Llama 3 3B (local)**, LLM APIs (Gemini, GPT, Claude), Prompt Engineering
– **LLM-as-Judge Evaluation**, Rubric Scoring, Retrieval Precision, Human-in-the-Loop Review Design
– QLoRA / PEFT, 4-bit Quantization (Unsloth), Vision-Language Models, HuggingFace, ONNX Runtime

Solution Architecture & Integration
– REST API Design, Elasticsearch Client Integration, FastAPI, Microsoft Graph API (OAuth2)
– Multi-Stage Retrieval Pipelines, Document Ingestion (PDF/Word, EN/CN/JP), Markitdown Extraction
– Data Flow Design, On-Premises Deployment, SQLite, Automated Reporting Pipelines

Machine Learning
– **PyTorch**, MobileNetV3, Transfer Learning, Dual-Head Architectures, Colab GPU Training
– Model Evaluation (F1, Accuracy, Precision), Failure Mode Analysis, Dataset Curation

Programming & Data
– **Python**, SQL, TypeScript, Dart, JavaScript, Bash, Git
– Grafana, Large-Scale Pipeline Monitoring, SQL Data Quality Validation, AWS (S3, Cost Explorer)

Documentation & Enablement
– Technical Documentation, Internal User Guidelines, Workshop Design & Delivery
– Stakeholder Briefings for Non-Technical Audiences, Cost-Benefit Analysis, Curriculum Authoring

Work & Research Experience

On-Premises LLM Document Automation & Evaluation — May 2026 – Aug 2026
AI & Software Engineer Intern, SUSS Digital Solutions & AI

- Built an on-premises document pipeline on a locally hosted **Llama 3 3B**, chosen so invoice data never left the university network; cut a 10-minute manual workflow to one-click review across 50+ documents.
- Designed an evaluation harness for lecture diagnostics: faster-whisper transcription into LLM-scored rubrics, plus a composite metric ranking delivery quality cohort-wide and a staff report rebuilt on each run.
- Automated lecture quality review on the **TwelveLabs** API, which judges engagement semantically rather than detecting silence, and returned trimmed footage plus flagged timestamps from 3-hour recordings.
- Co-presented AI tool evaluations and cost-benefit analysis to non-technical senior leadership, then translated the deployment decisions into business terms and answered their concerns about reliability and accuracy.

Production Retrieval System for an Internal Knowledge Base — Dec 2025 – Jan 2026
Machine Learning Intern, Yap Motor Singapore

- Built Yap Motor's first AI system, a production **RAG** pipeline (Google Gemini with ChromaDB) letting 24 mechanics query 500+ pages of repair manuals in real time; still in daily use six months later.
- Designed the retrieval pipeline in four stages: folder priority scoring, regex extraction of vehicle make/model/year from natural language, dynamic ChromaDB filters, then re-ranking before the LLM.
- Reached 90% retrieval precision across 8 diagnostic categories by tuning chunk size, overlap, and embedding strategy, validated over a month of mechanic feedback on whether results answered the query.
- Engineered the multilingual ingestion pipeline for EN/CN/JP PDFs and Word documents of varying structure; fixed two production blockers, ChromaDB schema validation failures and case-sensitive metadata filters.

Autonomous Document Triage Agent with Preference Memory — Jul 2026 – Present
Creator & Developer, Personal AI Project

- Built an autonomous email triage agent on the Microsoft Graph API that classifies and summarises high-volume university mail, using structured LLM extraction to score priority and draft per-thread action items.
- Added a persistent preference memory store, SQLite for feedback history and ChromaDB embeddings for similarity, so the agent re-ranks by cosine distance against what the user has previously marked useful.

- Automated the daily digest as a Jinja2 static dashboard pushed to GitHub Pages through the GitHub REST API, so each morning's summary publishes itself without anyone opening the tool or running a script.

Human-in-the-Loop LLM Translation Interface May 2025 – Aug 2025
Research Assistant, NTU Turing AI Scholars Programme (TAISP)

- Implemented bidirectional generation for a computer-assisted translation prototype: forward source-to-target suggestions, and backward refinement where an edit updates preceding suggestions in real time.
- Co-designed the search-based approach behind it, replacing single-suggestion machine translation with multi-option, translator-in-the-loop generation; presented as a poster supervised by Dr. Grandee Lee.
- Added probability-based alignment between source and target text at token and phrase level, so a translator could accept or reject a suggestion at sub-sentence granularity instead of sentence by sentence.

Enterprise Data Pipeline Monitoring & Integration May 2023 – May 2024
Data Engineer Intern, DBS Bank Singapore

- Built a Grafana and SQL dashboard monitoring a 3,000+ daily job production pipeline, surfacing version-specific bottlenecks that the owning teams then fixed, resulting in 20% less RAM and 15% faster runs.
- Developed an Elasticsearch REST client and integrated it into the bank's internal Python library, giving the team programmatic real-time feature retrieval for production model inference for the first time.
- Contributed to a platform migration as one of the team, integrating Kafka, S3, Yarn, and RabbitMQ components behind automated health checks and alerting that cut the rate of undetected pipeline failures.

Published Optical Music Recognition Model Jun 2026 – Present
Creator & Developer, Drum Hub

- Trained a dual-head MobileNetV3 model (2.47M parameters) for optical music recognition via two-phase transfer learning; 90.9% duration accuracy and F1 above 0.89, weights published on HuggingFace.
- Exported the trained model to ONNX Runtime so recognition runs fully offline on a local M1 Mac with no cloud dependency; documented the failure modes, including tom confusion and class imbalance.

Education

Nanyang Technological University (NTU) Aug 2024 – Dec 2028
Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

- Advanced Coursework: AI Fundamentals, Calculus, Linear Algebra, Probability & Statistics

Leadership & Activities

TAISP x MLDA Workshop Lead, NTU School of Computing 2025
Designed and led a Linear Regression and Neural Networks workshop for **207 participants**; built all slides, annotated Jupyter notebooks, and graded exercises from scratch, with a 100% completion rate.